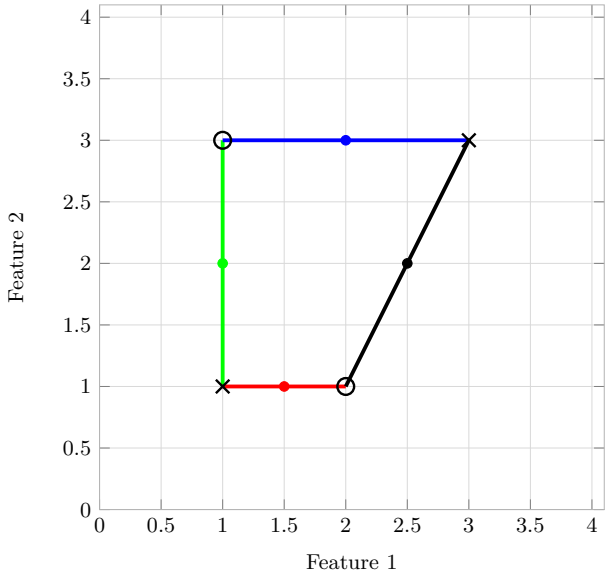
Finding the Decision Boundary of a 1-NN Classifier for Binary Classification

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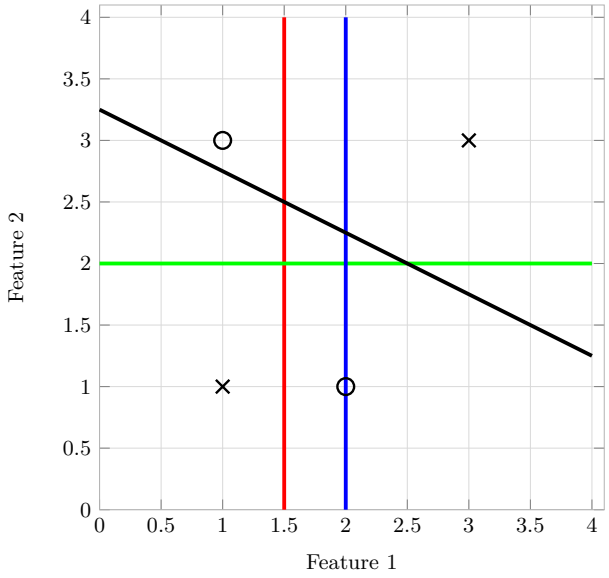
A decision boundary consists of points at which a classifier has no preference for one class over another. For a binary classification problem, the decision boundary consists of points that could be classified into either class. Since a 1-NN classifier assigns each test point the label of its nearest neighbor, a point could be classified into either class if it has two (equally distant) nearest neighbors that belong to different classes. Therefore, to find the decision boundaries of a 1-NN classifier for binary classification problem, we need to identify all the points that have two nearest neighbors belonging to two different classes.

To achieve this goal, follow the three steps below.

1. Identify the line segment connecting two data points belonging to different classes, and find its midpoint.



2. Draw a line perpendicular to the line segment at that midpoint.



3. Keep only the portions of these perpendicular lines where each point has two nearest neighbors belonging to two different classes.

